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Capstone - Final Report:

SUSHI ASSEMBLY LINE

PROCTECH 4TR1– C03

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Project Update



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Abstract

Our capstone project is a sushi roll production machine. The general issue we wish to address is that the standard sushi roll—although very popular around the world and relatively simple in construction—takes a disproportionate amount of time to make. Our device aims to drastically reduce the time required to prepare these simpler rolls.

The basic assembly of a sushi roll can be broken down into five simple steps: (1) lay out an even sheet of nori and rice; (2) add the roll's ingredients in a line across the sheet; (3) roll the whole assembly into an even cylinder; (4) slice the roll into pieces of even thickness; and (5) plate the pieces. Our device will have a designated station and mechanism for each of these steps. It will also feature a platform that moves from side to side to safely and precisely move the roll through the machine's processes. The design will emphasize safety features to prevent harm to the user and will be entirely food-grade.

This proposed solution supports chefs by automating repetitive preparation steps, freeing them to focus on higher-value culinary tasks. If successfully implemented, this system could serve both as an operational aid in restaurants and as a prototype for larger-scale food manufacturing applications.

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1. Executive Summary

Sushi is one of the most popular foods in the world, yet even the simplest maki rolls require significant time and repetitive manual effort to produce. During peak service hours, skilled chefs must devote large portions of their time to routine assembly tasks rather than higher-value culinary work for the more complex or artistic rolls. In grocery and food manufacturing settings, slow manual production limits output capacity and prevents businesses from scaling to meet growing consumer demand for ready-to-eat sushi products. This bottleneck drives up labour costs and creates operational strain across both industries.

The primary goal of this project is to design, build, and program an automated sushi assembly line at the scale of a medium-sized table that takes a sheet of nori from start to finish to create an entire sushi roll. This includes dispensing rice, laying toppings, rolling, cutting, and plating the result, without requiring human intervention beyond loading ingredients. The system is intended to produce uniform, food-safe sushi rolls consistently and efficiently, and to serve as a small-scale prototype for larger industrial applications. Secondary objectives include providing an HMI for customers to choose which toppings to add to create custom sushi rolls, ensuring all food-contact surfaces are food-grade, ensuring the design is modular so parts are easy to clean, and incorporating safety features to protect users.

The design was broken into six modular substations, each responsible for one step of the production process. A 1 meter aluminum v-slot linear guide rail (Station 0) serves as the backbone of the system, carrying a platform from station to station using a stepper motor and belt drive. Station 1 uses a pair of rolling pins to flatten a bed of rice onto the nori. Station 2 employs a dual-column compartment system with trapdoor mechanisms to drop pre-loaded toppings onto the rice in a precise line. Station 3 uses a “ribcage” rolling mechanism built into the moving platform, with two independently-driven stepper motors controlling opposing prongs to fold the nori around the ingredients into a tight roll. Station 4 lowers a frame of five parallel blades to cut the roll into six even pieces while the rolling mechanism holds it firm. Finally, Station 5 uses a UR3e robotic arm to pick and place the cut pieces onto a plate for serving.

By the end of the first term, the team successfully completed the design phase and initiated physical construction. The linear guide rail (Station 0) has been fully assembled, tested, and mounted to the project table, with limit switches at the ends and motor control confirmed as

functional. The latest version of the rolling mechanism (Station 3) has been fully 3D-printed. Both were demonstrated at the capstone showcase. Concept testing for the rice dispenser (Station 1) confirmed a 3mm rice sheet thickness and identified non-stick roller materials as a key area for further research and development. Design concepts for all remaining stations have been finalized and are ready for CAD modeling and part procurement over the summer.

If successfully completed, this system offers meaningful benefits to both restaurant and food manufacturing industries. In restaurant settings, the assembly line would take over the repetitive production of simple maki rolls, freeing skilled chefs to focus on more complex, high-value dishes. In a manufacturing context, it provides a proof-of-concept for scalable, automated sushi production to supply grocery retailers. The modular station design makes the system easy to service, clean, and adapt, and the use of food-grade materials and safety guards ensures it meets the standards required for real-world food service environments. The goal is to gain valuable experience in the engineering design process from the iterative feedback of our mistakes and troubleshooting in an automation context.

Several challenges have already been encountered and others are anticipated. Initial testing of the rice dispensing station showed that rice adheres to silicone rolling pins rather than releasing cleanly, requiring further materials research. The rolling mechanism required a design change from a single synchronized motor to two independent stepper motors after testing revealed that simultaneous folding caused the nori seaweed to collide at the top and prevent the roll from closing properly. Looking ahead, sourcing food-grade custom components that meet all design specifications is expected to be a significant challenge. Integration of all six stations into a cohesive, synchronized system will also require substantial programming and debugging effort from the team in the second term.

Over the summer, the team plans to complete all CAD models and electrical diagrams, finalize the Bill of Materials, and order all components to allow maximum build time in the fall semester. Priority areas for future development include completing the rice and ingredient dispensers, refining the cutting mechanism with angled blades to improve slicing performance, selecting and integrating an appropriate end-effector for the UR3e robotic arm, and programming the full state-machine control logic to coordinate all six stations. If time permits, the team also intends to implement an HMI for ingredient selection. In the longer term, this design could serve as the foundation for a larger-scale industrial sushi production system.

2. Introduction

2.1. Motivation

Sushi is a beloved food all around the world, as sushi restaurants are anticipated to grow their global market size by $3.92 million USD growth in the next three years [1]. As such, sushi restaurants and grocery retailers face a consistent operational challenge: basic sushi rolls are in extremely high demand [2], yet they require significant time and repetitive manual effort to produce. Although these maki rolls are simple in design, the preparation process involves multiple sequential steps that must be performed carefully and consistently. This creates a bottleneck in both restaurant kitchens and large-scale grocery production settings.

The problem manifests clearly during peak service hours and high-volume production periods. Orders accumulate, preparation times increase, and skilled chefs must devote large portions of their time to repetitive tasks rather than higher-value culinary work. In grocery supply chains, slow production limits output capacity and reduces the ability to meet growing consumer demand for ready-to-eat sushi products.

This inefficiency leads to several difficulties. Restaurants may experience longer wait times and operational strain, while grocery producers face constraints in scaling output. Labour costs increase when highly trained chefs spend time on routine assembly rather than specialized dishes. Over time, this imbalance reduces productivity, increases pressure on staff, and limits overall growth potential.

At the beginning of this project, our team expected to explore whether automation could meaningfully reduce these inefficiencies without compromising quality. We aimed to design a solution that would streamline production, improve consistency, and support both restaurant operations and larger industrial environments.

To address this need, our project proposes a sushi assembly line concept that automates repetitive preparation tasks while maintaining product quality and consistency. At a high level, the system is designed to support chefs rather than replace them, freeing their time for complex creations while enabling scalable production for grocery distribution. The goal is to increase efficiency, reduce strain on skilled labour, and create a more sustainable production model for

high-demand sushi products. The system will also be a prototype for a large-scale assembly line for food plants.

2.2. Project Objectives

The proposed objectives for this automated sushi assembly line are as follows:

- It shall produce uniform pieces of sushi that are consistent in appearance and quality
- It shall not require human intervention to complete the process, aside from replenishing the ingredients/materials
- It shall be safe for humans to interact with the system if need be
- It shall be entirely food-grade and allow for safe food consumption
- It shall have an HMI to display relevant data (ex. current station, runtime data, etc.)
- It shall be quick and efficient

2.3. Project Requirements

The proposed requirements for this automated sushi assembly line can be broken down into each individual station and are as follows:

2.3.1. Station 0: Linear Guide Rail

- It shall be 1 meter long and fit atop the surface of the provided table
- It shall attach to the table using at least two M8 screws
- It shall have v-slots to allow the bearings to roll across the bar

2.3.2. Station 1: Dispensing the Rice

- It shall have two rollers placed 4mm apart
- It shall have a maximum length of 1 foot for each rolling pin
- It shall be driven by a single 8-35V stepper motor
- It shall be made of a non-stick material (ex. wood)

2.3.3. Station 2: Laying the Toppings

- It shall have a 2x4 array of compartments, each with a trap door that opens to drop its ingredients
- It shall be driven by two 8-35V stepper motors

- It shall be made of food-grade plastic (ex. polypropylene, high density polyethylene, etc.)
- It shall have a maximum length of 15cm (the length of the nori and moving platform)

2.3.4. Station 3: Rolling the Sushi

- It shall have a flat platform roughly 15cm x 13.5cm for the nori, rice, and toppings to be placed on
- It shall carry two 8-35V stepper motors that drive each individual 'claw' of the folding mechanism

2.3.5. Station 4: Cutting the Roll

- It shall have 5 blades that cut the roll into 6 pieces of uniform thickness
- It shall cut the roll while it is still being held by the 'claws' in the previous station, cutting between each part of the claw
- It shall be driven by a servo motor in order to produce more force

2.3.6. Station 5: Plating the Pieces

- It shall employ a rubber end-effector to allow for pick-and-place operations

3. Literature Review

Automation of sushi preparation is not a brand-new concept. There exists commercial machines designed to perform specific sushi-making tasks that have been adopted in some restaurants and food service environments. A study from 2025 states that robotic sushi-making systems have achieved a 12% adoption rate across chain restaurants in some form [2]. A notable example in this field is the lineup of commercial sushi machines from AUTEC.



Figure 1: *A rice roller machine from AUTEC.*
By https://www.sushimachines.com/technical-support/

AUTEC is a company with a long history in sushi automation tools dating back to the 1980s when they released some of the first commercial sushi forming devices and have continued expanding their product range globally [3].

The issue is that they are focused on individual production steps such as creating rice sheets or cutting rolls. This is the current state of sushi machines, or the “prior work” we are building on.



Figure 2: *UR3 Universal Robot making coffee.*
Screenshot taken from https://www.youtube.com/watch?v=t83wtiVurUI

This project from Mobile Automation demonstrates the capabilities of the UR3 Universal robotic arm as it is used to make a cup of coffee [4]. This provides precedence for how our group intends to use the robotic arm in the last station to pick up the sushi pieces and place them on a plate. Projects like this prove that the UR3 lineup of products can be used for much more complex tasks than our group plans to use it for. However, the use of a robotic arm is simply one part of our intended design.

The gap this project seeks to build on is to move beyond isolated machines that automate *parts* of sushi preparation toward a more integrated assembly line approach. Instead of simply producing rice balls or singular rolls, the proposed system aims to connect and coordinate tasks into a streamlined workflow, reducing the manual intervention still present in existing machines and supporting high-volume production for markets like restaurants and grocery distribution. Our proposed solution will include a full assembly line for a sushi roll from scratch, with multiple substations on a small table with wheels for placing ingredients, rolling the sushi, cutting, and placing them on a plate, ready to serve.

4. Design Concept

4.1. Alternatives & Proposed Solutions

Like any good assembly line, this project can be broken up into the following workstations:

- *Station 0:* Linear Guide Rail
- *Station 1:* Dispensing the Rice
- *Station 2:* Laying Out the Toppings
- *Station 3:* Rolling the Sushi
- *Station 4:* Cutting the Roll
- *Station 5:* Plating the Pieces

Each station is being designed independently, with the outputs of one station being expected to become the inputs of the subsequent station. This encourages a seamless integration of workstations when all is put together.

Function	Option 1	Option 2	Option 3
Foundation	Conveyor Belt (Continuous Process)	Linear Guide Rail (Batch Process)	
Preliminary Set-Up	Place Nori in Center of Bamboo Mat; Select Type of Sushi Roll; Place Plate at End; Press Start;	Place Nori in Center of Platform; Select Type of Sushi Roll; Place Plate at End; Press Start	Place Nori in Center of Bamboo Mat; Place Plate at End; Press Start (No Sushi Type Selection)
Station 1: Flattening the Rice	Hopper of Rice; 4 Wooden Rollers; Machine Moves Over Platform	Hopper of Rice; 2 Wooden Rollers; Platform Moves Under Machine	Hopper of Rice; 2 Silicon Rollers; Platform Moves Under Machine
Station 2: Dispensing the Neta	Single Column with Individual Compartments; Platform Moves Under Machine; Compartments Rotate to Drop Ingredients One By One	Single Wide Column with Individual Compartments; Platform Stops Under Machine; Floor of Each Compartment Opens to Drop All Ingredients At Once	Two Single Wide Columns with Individual Compartments - One for Fish, One for Avocado; Platform Stops Under Machine; Floor of Each Compartment Opens to Drop All Ingredients At Once
Station 3: Rolling the Sushi	Prongs That Protrude From Moving Platform; Grips Nori From Bottom; Tightens Using Stepper Motor	Stationary Prongs; Platform Moves Under Machine; Scoops Bamboo Mat and Roll From Above; Tightens Using Stepper Motor	Stationary Machine Attaches to One Edge of Bamboo Mat; Folds Over the Roll; Tightens Using Stepper Motor
Station 4: Cutting the Roll	Series of 5 Parallel Straight Blades; Machine Comes Down Onto Bamboo Mat and Presses; Plastic Guards Surround Blades and Retract When Pressed	Series of 5 Parallel Straight Blades; Machine Comes Down In Front of Bamboo Mat and Moves Back to Cut Roll; Large Plastic Guards Encapsulate Roll and Blades	Series of 5 Parallel Seraded Blades; Machine Performs Sawing Motion to Cut Roll; Large Plastic Guards Encapsulate Roll and Blades
Station 5: Plating the Pieces	Pick-and-Place End Effector; Camera Vision to Locate Sushi Pieces	Pick-and-Place End Effector; Hardcoded Pick-Up and Drop-Off Locations	

***Table 1**: Morphological chart with options for each assembly line substation.*

Regarding the foundation of the system, our group had to decide between using a conveyor belt or a linear guide rail to transport the unfinished roll from station to station. The former lends itself to continuous processes due to being able to have multiple rolls placed on it, while the latter suggests a continuous process as only a single roll will be carried from end-to-end. Our group ultimately decided on using a linear guide rail system as this would emphasize quality and consistency in our rolls, over efficiency. Having a single platform that carries the roll from station to station also allows for more flexibility, and the guide rails allow for easy installation of limit switches which will tell the microcontrollers exactly where the platform is located.

Before the automated line begins, certain tasks need to be completed by the operator/chef. A sheet of nori should be placed in the center of the platform and a plate should be placed at the end of the assembly line. Finally, the operator should select the desired neta (fillings) for the sushi roll and then press START on the HMI. Occasionally, the operator will need to replenish the ingredients in the compartments in Station 2 as ingredients like fish and avocado should not be left out at room temperature for longer than two hours [5].

With the assembly line running, the platform stops at Station 1: Flattening the Rice. The goal of this station is to dispense a bed of rice neatly onto the sheet of nori. This machine has a hopper of cooked sushi rice at the top, which is fed into a pair of wooden rollers that flatten the rice to a certain thickness. We decided that having the platform and sheet of nori slowly move forward under the machine would be the best way to achieve a uniform bed of rice, as opposed to having the machine move over a stationary platform.

Next, the platform will stop at Station 2: Dispensing the Neta. This station intends to lay out the toppings (fish or avocado) on the bed of rice in a manner that allows everything to be rolled together in the next step. Initial designs used a single column of fish, where a vertical conveyor belt system would drop one piece of fish as the platform positioned itself below. To simplify this further, we widened each compartment in the vertical column so that the machine just had to drop its floor in order to release the ingredients held inside the bottom compartment. This way, the platform could be positioned directly below the machine and the fish could be dropped onto it all at once in the desired position. With this vertical column, it acts as a queueing system for the toppings and the upcoming rolls. Within the stack of compartments, the bottom one releases its contents onto the rice and then closes, and then each of the upper compartments open in sequence to move the desired toppings down. Our final iteration simply adds a duplicate vertical column side-by-side with the current column in order to allow for more variety in ingredients and better placements (ex. avocado should be placed in the compartment at the same level of the fish, not on top of it).

After many design iterations for Station 3: Rolling the Sushi, we found the most success having the rolling mechanism built into the platform itself. Upon leaving Station 2, curled prongs hidden beneath the platform will reach up to grip the bamboo mat along with the sushi ingredients inside. By applying force with the stepper motor, the goal is for the ingredients to press in on itself and form a tight roll. Our latest iteration sees the use of two separate motors to control each individual prong. Instead of folding the entire roll at once, this allows one prong to roll the sushi on one side and then fold the nori over on the other side which more closely resembles how humans perform the operation.

Station 4: Cutting the Roll uses a row of 5 blades in parallel that will come down to slice the roll into 6 pieces. We have decided to use straight blades that will come directly down onto the roll, but real-world tests will have to be performed to ensure that this is the best method of slicing. Nonetheless, safety is at the forefront in this station so clear plastic guards will be put around the blades so that humans do not have easy access. The guards will encapsulate the entirety of the sushi roll and retract when pressed down into the plate while slicing.

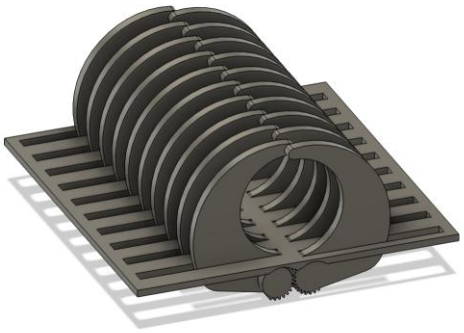


Figure 3: *Preliminary design for Station 3: Rolling the Sushi*

Finally, the platform will move to Station 5: Plating the Pieces. A UR3e robotic arm will be used as a pick-and-place robot, and the appropriate gripper end-effector will be attached. Ideally, the location of the sushi pickup and drop off points will be hardcoded into the robot. However, if this proves to be unfeasible during the execution stage, camera vision and processing will have to be considered.

4.2. Bill of Materials

Since we are halfway through our project and we have another semester to complete it, this BOM is a work-in-progress list of parts that we have finalized, ordered, or used so far. The complete BOM will be finished next semester.

Category	Part Number	Qty	Description	Purchase Source/Link	Unit Price
[Mechanical] Station 0: Linear Slide Base					
Mechanical		1	1m 40x40 Aluminum V-slot Rail	Link	$73.91
Mechanical		1	3D Printer Belt 5 Meters (16.4 ft)	Link	$9.59
Mechanical		2	Non-stick bamboo mat	Amazon	$4.50
Mechanical	625ZZ	4	Linear bearing	On Hand	On Hand
Mechanical		4	POM Wheel	On Hand	On Hand
[Mechanical] Station 1: Rice Dispenser					
Mechanical		2	Plastic rolling pin with silicon layer	Dollarama	$5.00
[Mechanical] Station 4: Cutter					
Mechanical	GRA-EXC 20021	5	Straight blades	Model Kits Canada	$4.95
[Mechanical] Station 5: Plating Robot					
Mechanical		1	UR3e robot	On hand	On hand
Mechanical		1	Plate	On hand	On hand
[Electrical] Station 0: Linear Slide Base					
Electrical		1	Arduino MEGA		
Electrical	(Creality 42-32) JK42HS34-0844YA-01	1	Stepper motor	On Hand	On Hand
Electrical	L298N	1	Stepper motor driver	Amazon	$3.50
Electrical	ME-8112	5	Roller plunger limit switch	AliExpress	$3.77
Electrical	KW10	2	Limit switch	On Hand	On Hand
[Electrical] Station 1: Rice Dispenser					
Electrical	(Creality 42-40) JK42HS34-0844YA-01	1	Stepper motor	On Hand	On Hand
Electrical	L298N	1	Stepper motor driver	Amazon	$3.50
[Electrical] Station 2: Fish Dispenser					
Electrical	(Creality 42-34) JK42HS34-0844YA-01	1	Stepper motor	On Hand	On Hand
Electrical	L298N	1	Stepper motor driver	Amazon	$3.50
[Electrical] Station 3: Ribcage Roller					

Electrical	(Creality 42-34) JK42HS34-0844YA-01	1	Stepper motor	On Hand	On Hand
Electrical	L298N	1	Stepper motor driver	Amazon	$3.50
[Electrical] Station 4: Cutter					
Electrical		1	Servo motor		
[Miscellaneous] Station 0: Linear Slide Base					
Miscellane-ous			Plastic gloves		

5. Completed Work

This term we worked hard every week to complete as much of our project as possible. We decided within the first two weeks that we wanted to divide the complete device into a number of subsystems, each of which will complete a step of the process. This batch processing will allow us to focus on one roll at a time. By week two we had decided what tasks we wanted to create subsystems for. We also decided that we wanted to use a linear rail system to carry the roll between stations.

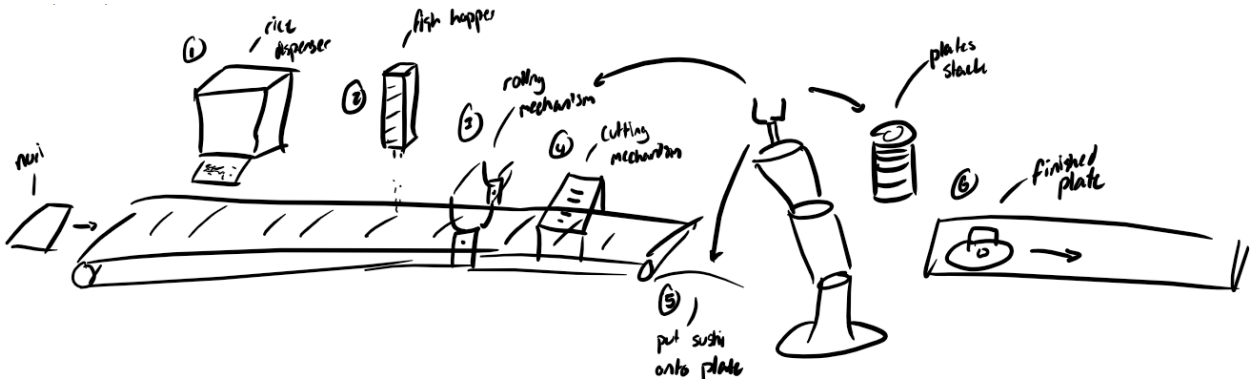


***Figure 4**: Rough concept sketch for our project.*

Because our project can be divided into 5 subsystems, each of which can be independently designed and developed, we will individually look at each system's completed design and completed work thus far.

5.1. Station 0: *Linear Guide Rail*

As the foundation of the entire project, we decided to begin with Station 0 first. We happened to have a spare, non-functional Ender 2 3D printer on hand that we dismantled to repurpose the motors, early in the process. We were therefore also able to use some of the functionality of the 3D printer as inspiration for our project. The design we chose for linear guide platform was inspired by the z-axis of this 3D printer. The bearings, nuts & bolts, motor mount, limit switch, belt tensioner and motor were all repurposed and used in our final version. However, the biggest complication came with upscaling this rail to span a ~1 meter long table.



***Figure 5**: Ender 3 v2 3D printer.*
***By** https://www.creality.com/products/ender-3-v2-3d-printer-csco*

In week 8 of our process, we purchased a one meter long aluminum v-slot extrusion rail and a 2 meter long belt. We were originally hoping to use some of the aluminum rails that were already vacant in the capstone room; however, after further investigation we found out that these rails are T-slot, meaning that they will not fit the bearings we were planning on using.

While waiting for the order to arrive we purchased an A4988 stepper motor driver and used it to test the functionality of our stepper motor on the shorter linear rail system from the 3D printer.

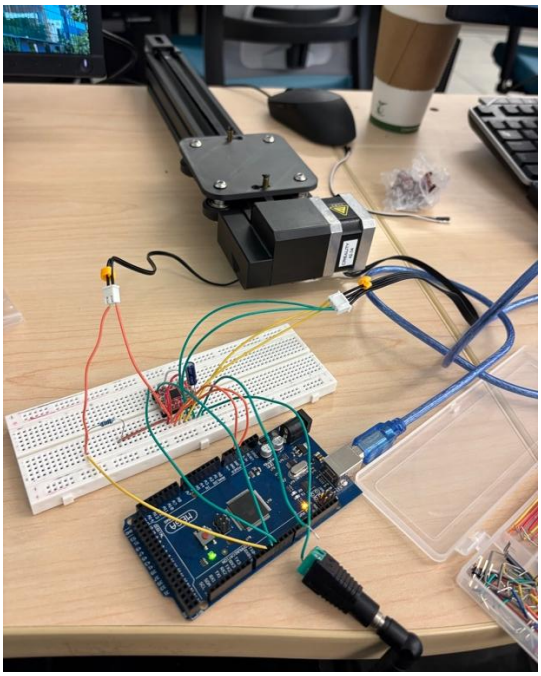


***Figure 6:** Testing our linear rail and motor driver.*

By week 10, we finally received the belt and rail and were able to assemble the full subsystem. The following week, we ran another test with the motor driver to prove that the system was fully functional. We also drilled two holes through the aluminum rail so that the whole system could be firmly fastened to the table we plan to mount all of our systems on.

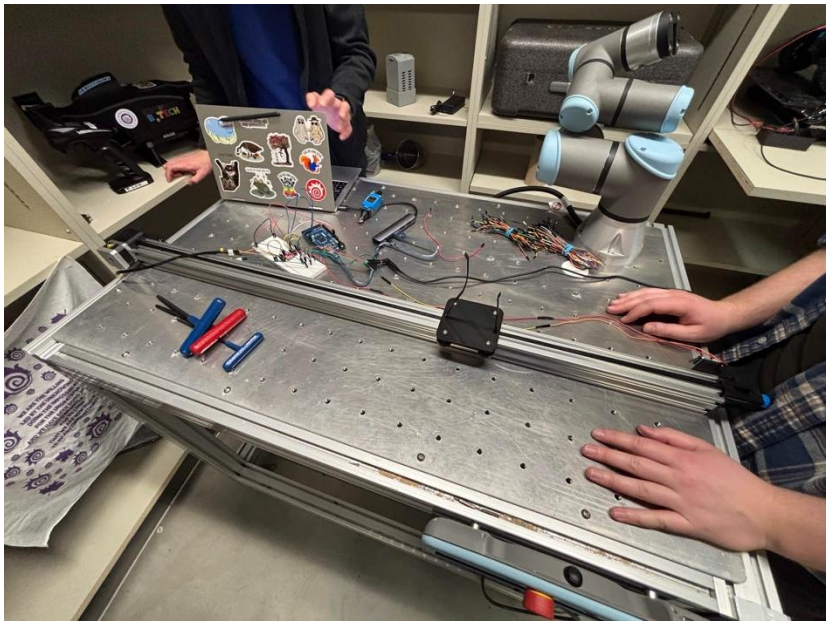


Figure 7: *Full test of the 1meter rail.*

5.2. Station 1: *Rolling the Rice*

The rice rolling station has been one of the most technically challenging ones we have come across thus far. As previously mentioned, our design has been inspired by the AUTEC rice machine. The initial design was originally conceived in week 3, discussed and varied through the following few weeks and the final design concept was finalized and sketched by week 6.

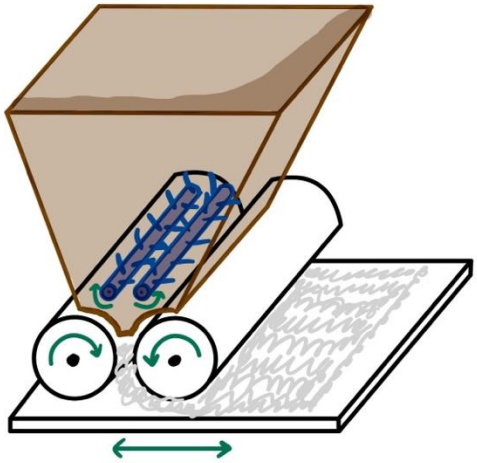


Figure 8: *Initial design sketch of the rice dispenser.*

The main features we decided we need for this module are two large rollers for compressing the rice, a large hopper that can store all of the rice stock and agitator arms that will churn and aerate the rice as well as pushing it towards the rollers so that we won't need to rely on it being gravity-fed. We also decided that we want this device to be mounted just above the moving platform and to have the platform move below past the dispensers output at the same speed as it's dispensing in order to lay out a perfectly even sheet.

Our main concern with this device would be that the rice would stick to the rollers instead of being dropped out from between them. In order to resolve this we chose to purchase two silicone rolling pins, hoping that the rice would be unable to adhere to the silicone. Therefore we chose to bring a batch of rice for weeks 8 and 9 in order to test this system.



Figure 9: *Testing our rice rolling system.*

These tests confirmed our fears that the rice would stick to the rollers, and we were not able to dispense a perfect sheet.

These tests also allowed us to determine the distance between the two rollers, and therefore the thickness of the rice sheet, that we wanted to use in the final version. As you can see in *Figure 11*, our testing jig involved a number of differently spaced frames to hold the two pins at different distances and also involved a version where the pins were mounted to gears so that they would only spin in synchronous with the other. Ultimately, we decided that a thickness of about 3 mm is best and that we will need to do further research and tests as to the best option for a non-stick rolling pin.

5.3. Station 2: *Laying the Toppings*

The mechanism for dispensing the ingredients has been one of the hardest to decide upon through this project. The concept of the subsystem isn't too complicated – simply drop the ingredients the platform is underneath. However, this device also needs to be able to drop the ingredients in a perfect line perpendicular to the rice sheet, have the ability to drop different ingredients based on the selected ingredients, have storage space to stock multiple different ingredient types, and still be completely food safe.

The first design version we introduced (*Figure 9*) was essentially a large wheel that would drop the ingredient as it gradually spins. However, the concerns we had with this version are that we would need multiple in series to dispense a full line, it doesn't have very much storage for the backlog and doesn't have any variability for different ingredients.

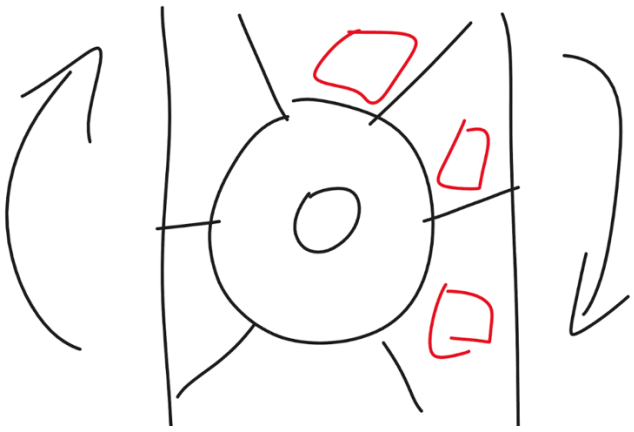


Figure 10: *Initial concept for the ingredient dispenser*

For the next design iteration, we decided to rely on a series of compartments with trap-door mechanisms for dropping the ingredients. These compartments will be as long as the roll so that when the ingredients are dispensed, they will already be in a perfect line at the correct length. By having two columns of compartments side-by-side we are also able to drop two different ingredients at the same time. All the compartments above the bottom two can now be used as a queuing system to lower the ingredients one by one towards their release, giving us the required storage for the backlog. Finally, this system can be made of completely food-safe materials. By week 6 we created a sketch of this final design concept (*Figure 10*).

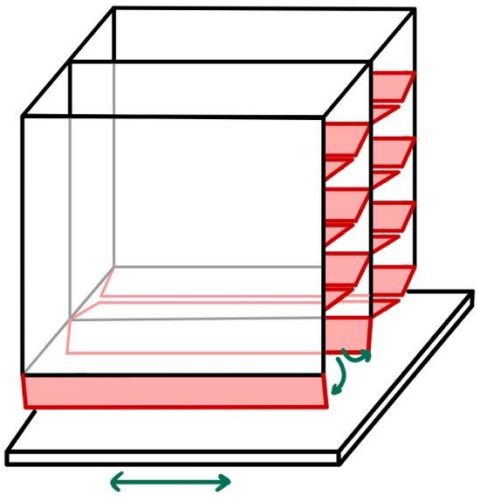


Figure 11: *Final design concept for the ingredient dispenser.*

We have not started the assembly of this system yet but plan to have a completed CAD model ready by the end of summer as to begin assembly in the fall term.

5.4. Station *3: Rolling the Sushi*

The rolling system is the next most important subsystem that we have been most of our effort into. This mechanism is meant to roll the sushi tight together before it can be cut and plated. This is crucial as if the sushi is not tightly rolled together, it will fall apart upon cutting and plating.

The design for the rolling mechanism was first conceptualized in week 2 and has undergone many iterations since.

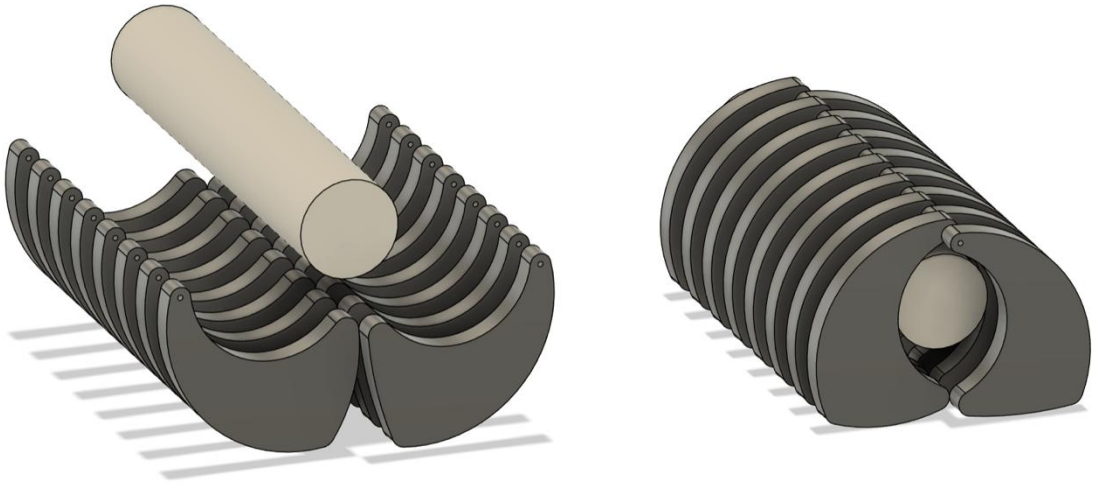


Figure 12: *Initial design concept from week 2.*

This first concept built off the idea of holding each side of the bamboo mat and pulling them up and around the roll in order for it to be compressed from all angles. In weeks six, seven, and eight we designed a small version of this mechanism and tested it with real nori seaweed and sticky rice.



Figure 13: *First test using the rolling mechanism.*

We discovered in testing this version that synchronizing the two sides of the mechanism would result in the seaweed colliding at the top of the roll interfering with its ability to completely close around itself. Therefore, for the next version of the roll we decided to individually power the two different sides of the roller with two different stepper motors.

We also determined that it would be too difficult to move the partially completed roll onto and off of the rolling mechanism and therefore that the entire mechanism should be mounted atop the previous linear guide rail and move through the system with the sushi.

Both of these decisions resulted in the newest version of the rolling mechanism. This version also addresses the placement of the two stepper motors which will be mounted to the sides of the linear platform and connect to the two sides of the rolling mechanism with use of gears.

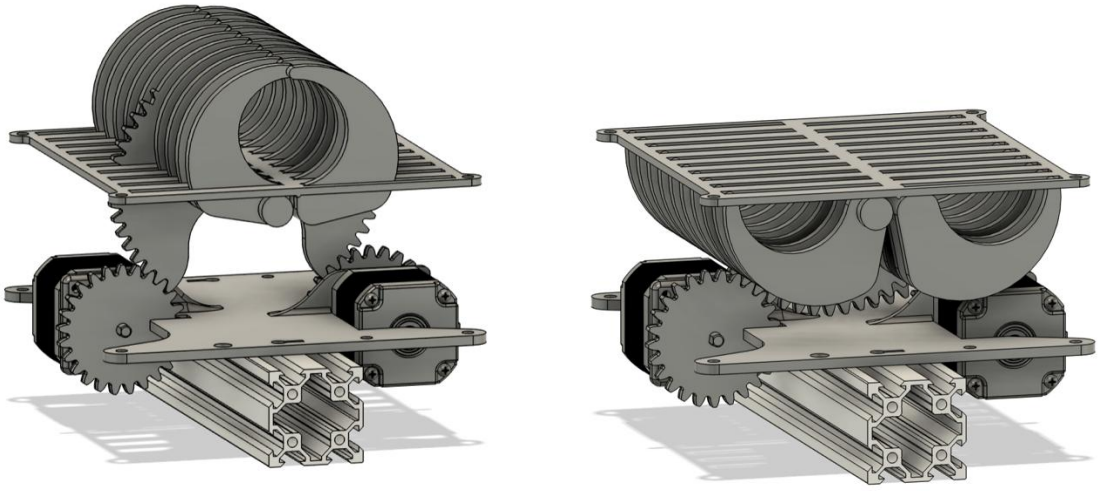


Figure 14: *Current version of the rolling mechanism and linear rail mount.*

This version we completely 3D printed and brought to our capstone poster presentation as a demonstration.

5.5. Station 4: *Cutting the Roll*

The next station is meant to cut the roll into 6 even pieces of sushi. The concept for this station is relatively simple and has not undergone much iteration since its initial conception in Week 2. We plan to place 5 blades of equal length and equal spacing in a frame so that they are

parallel with one another. This frame will then lower over the rolls allowing each of the blades to slice through the roll.

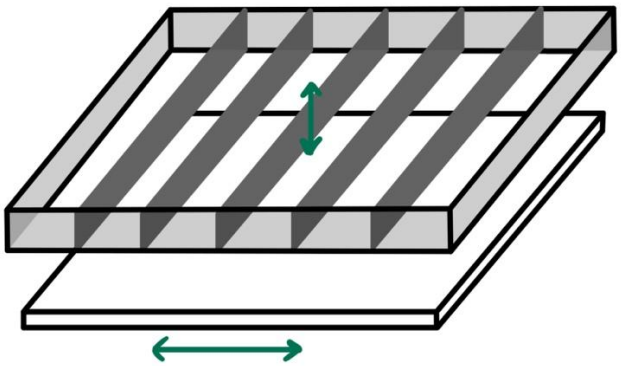


Figure 15: *Sketch of the cutting mechanism.*

Our main concern with this mechanism is that upon lowering the blades over the roll it won't cut them so much as it will compress the entire roll. To solve this, we are hoping to combine the rolling and the cutting mechanisms. By holding the roll tightly within the teeth of the rolling mechanism while lowering the blades between them, it will hold the roll tight while it is cut, preventing it from being merely crushed. We were also inspired by the blade of a guillotine, which is at a slanted angle to cause more of a slicing than a pressing motion. We hope to use this technique as well. All these concepts will need to be designed over the summer break and tested early in the fall term.

5.6. Station 5: *Plating the Sushi*

For our final station we plan to use a robotic arm for plating the sushi pieces. Our professor gave us access to a UR robotic arm which we will program to individual pick up each piece of sushi and place them on the final plate. The table that we plan to mount all of our substations to was also provided with the arm, which means that it can be easily mounted to the surface.

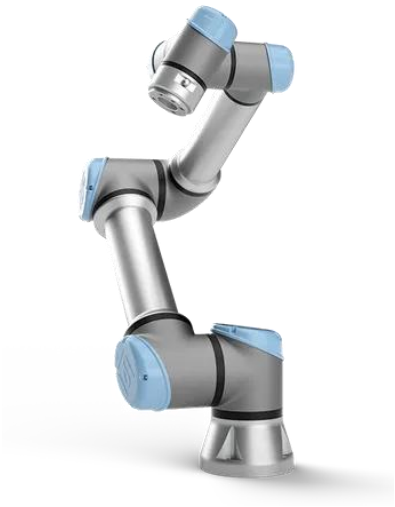


Figure 16: *Universal Robots – robotic arm*

We do not have an end effector for this arm yet but we plan to purchase a gripper from UR if possible, or design our own once we have the sushi rolls to the point where we can test the arm on them.

6. Discussion

From the initial proposal phase of the project, the following timeline was put forward:

- *Weeks 1-3:* Brainstorm, assess, and finalize project ideas; create initial concepts
- *Week 4:* Present proposal
- *Weeks 5-6:* Narrow down design concepts; develop design prototypes; begin concept testing
- *Week 7:* Midterm report and presentation
- *Weeks 8-10:* Form BOMs, order key parts, build prototypes of key components
- *Week 11:* Create poster for presentation
- *Week 12:* Finalize work for the term; write final report
- *Week 13:* Poster presentations

Based on this, our group has successfully met the deliverables of each week which puts us right on track for the end of the first semester. Moving forward, the focus will be on purchasing parts, constructing the machines, programming the microcontrollers, and debugging the system.

By the start of the next term, we would like to have all our materials and planning completed so that we are ready to start building the project. This includes the finalized designs of each station, complete with specifications and measurements in a CAD file, an electrical diagram showing the IO wiring for each microcontroller, and a fully fleshed-out state diagram. The plan is to source and order parts over the summer to allow the vendors time to ship, or potentially custom-build, our desired components. This would give us the entirety of the fall semester to put the project together and face any issues that arise during this phase.

We are purposely allowing ourselves the most time to plan, test, and generate prototypes because we recognize the importance of creating high-quality designs from the start. The time investment put into the design stage now will ensure that we do not spend as much time reworking or redesigning the project during the execution stage next semester.

While the project seems feasible based on our current progress, we are prepared to narrow the scope as a contingency plan in the event that certain aspects of the sushi assembly line cannot be achieved in time. For example, the initial proposal mentions using the HMI to select which

ingredients would go into the roll. If providing the option for multiple different toppings proves to be too difficult to implement into the program, we can scrap the idea without radically altering the design of the fish dispensing station. Additionally, a more drastic scope reduction would be to remove the entire automated section of flattening the rice. Instead, this would be a preliminary manual step for the chef to prepare beforehand.

Looking forward, a significant foreseeable issue stems from the ability to source the necessary parts in order to build each apparatus. With each complex, custom-designed machine, it may be difficult to find the right equipment that can meet all of our specifications as well as requirements (i.e. it must be food-grade, be able to be disassembled for cleaning, etc.). This is why we are aiming to source these parts early so that we know what is available and what is not, and we can consider alternative options such as custom manufacturing the parts or using a different material.

With existing technologies like the AUTEC rice roller machine, we can look to their designs as inspiration for specific sections of our assembly line process. For example, it is evident that the process of flattening and outputting the rice can be accomplished with only two motors and two rolling pins. However, our project aims to provide a seamless, end-to-end automated solution for creating sushi, which has not been accomplished at this scale before.

If successful, this project could see use in high-end sushi restaurants to be used alongside professional chefs. The sushi assembly line is meant to take the burden of creating the simple, yet high demand rolls off the chefs, allowing them to focus on the restaurant's more complex and impressive dishes. It can also act as a smaller-scale prototype for the mass production of prepackaged basic sushi rolls for sale in grocery stores.

7. Conclusion

The practical applications of an automated sushi assembly line machine cover two main industries: restaurants and food manufacturing. The first application involves sushi restaurants where chefs must make rolls to satisfy high demand quickly, and simpler rolls are a repetitive task that has an opportunity to be automated to free their time for the more complex, artsy rolls. The second application involves manufacturing off-the-shelf sushi packs to be sold in grocery stores, and large volumes of rolls must be continuously made or batch-processed to satisfy customer demand.

With this project, we aim to gain knowledge and skills in large-scale industrial automation that businesses use with a small-scale budget system that can produce small rolls at a time. We believe we will gain experience in mechanical and CAD design, assembly line control and device interfacing, and batch processing programming. The largest challenge will be to design the system mechanically and order all parts, as well as the inevitable troubleshooting that will arise in all aspects of the design process. We aim to learn principles of the engineering design process with iterative problem-solving and learning from mistakes to come up with a final design that finally satisfies all our requirements through loops of feedback.

Currently, our project progress aligns with our current objectives. We have completed most of the theoretical and conceptual design work, as well as initiated construction on the first base station—arguably the most important—and figured out how to mount it on our table. We believe as long as we keep up the progress, as well as put in some time for the project during the summer, we can achieve the objectives of our schedule smoothly. Our aim is to always work on the project as if it's due next week, and then use the remaining time we may have for the troubleshooting process and to improve on the design.

The functionality of the prototype—in this case, the tangible progress we have so far—currently meets the expectations of usability. The linear slide gets controlled by our limit switches and stepper motors as expected, and at a suitable speed.

We will continue making progress on the project as the weeks go by. The main things to focus on are the mechanical designs, the CAD files, and shopping for parts for the designs and filling out our BOM.

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